ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. Isolating specific categories allows us to focus on domain-relevant patterns, such as identifying frequently co-occurring keywords within a specific category.

AI used for this answer: Yes

If yes, how: brainstorming and grammar

External references (links):
https://rasbt.github.io/mlxtend/user_guide/frequent_patterns/apriori/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: If we discard all 40% of the data, we will lose too much information. An alternative is to estimate missing value of category_name by using text data to train a classification model.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The propotions do not perctly match the original even the distribution in the plot looks almost the same.
2. The sampled data will possibly have strong bias.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features increases from 35482 to 35788.
2. Adding stopwords back can make the matrix less sparse by increasing the number of non-zero entries. However, these common words might make it harder to identify important information and meaningful patterns.

AI used for this answer: Yes

If yes, how: Grammar

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The total size of vocabulary decreases as the variance threshold increases.
2. The noisy rare words such as *zipped*, *zimmerman*, *4202*.
3. It is generally safe since features with near-zero variance has almost the same values across documents, which means they are not the important information for distinguishing between different classes.

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: Grammar correction, concept clarification, logical refinement

External references (links): https://topepo.github.io/caret/pre-processing.html#zero--and-near-zero-variance-predictors

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Most of the words are domain-specific such as *graphics*, *vga*, *video*. But there are also some generic stop words like *thanks*, *writes*, *people*.
2. Pearson correlation is a measure of linear relationship using original values, while Spearman carrelation is a measure of monotonic relationship by ranking.
3. It happens when the relationship is monotonic but not linear, such as an exponential pattern. In a text mining context, this can occur when a word associated with a class appears an unusually large number of times in only one or a few documents. In that case, Pearson correlation may be lower because it is sensitive to large differences in raw values, while Spearman correlation remains high because the ranking is less affected by such outliers.

AI used for this answer: Yes

If yes, how: Grammar correction, concept clarification

External references (links):
https://www.sciencedirect.com/science/article/pii/S0098300411002251

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?

3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Run time increases from 2.94 s to 0.21 s, while pattern count increases from 710 to 9547.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

Top-K FAE keeps the top k patterns with the highest support counts, while MaxFPGrowth uses minSup as a minimum support threshold to keep maximal frequent patterns that meet this threshold.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they avoid retaining smaller redundant subset patterns that are already contained in larger frequent patterns.

AI used for this answer: Yes

If yes, how: Grammar correction, concept clarification

External references (links):

https://link.springer.com/article/10.1007/s41019-024-00241-2
https://michael.hahsler.net/arules/reference/is.maximal.html

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. It is impossible to visualize all the features directly because the dimensionality is too high, while a scatter plot can only display up to three spatial dimensions.
2. Dimensionality reduction algorithms.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. UMAP shows distinct, separated cluster islands, while PCA shows heavy overlap around the center**.**
2. It struggles to separate high-dimensional text categories since it finds directions of maximum variance and therefore affected by the outliers.
3. We can't over-interpret the physical distance in a t-SNE plot since it preserves local structure rather than global. The distance may be distorted during dimension reduction.

AI used for this answer: Yes

If yes, how: Grammar correction, concept clarification

External references (links): https://distill.pub/2016/misread-tsne/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 columns
2. They assume the raw integer column has order or meaningful distance.
3. It tralate the raw number into separate binary column, removing the false numerical order and distance relationships between categories.

AI used for this answer: Yes

If yes, how: Grammar

External references (links):
